

Multiple Choice (10 marks)

1. In Python 3, what is `['a', 'Chemistry', 0, 1][1]`?

- (a) a
- (b) Chemistry
- (c) 0
- (d) 1

2. In Python 3, what is the output of `print list[1:3]` if `list = [ 'abcd', 786 , 2.23, 'john', 70.2 ]`?

`786 , 2.23, 'john', 70.2`

- (a) abcd

`786, 2.23`

- (b) None of the above.

3. In Python 3, which of the following operator in python performs exponential (power) calculation on operands?

- (a) `**`
- (b) `//`
- (c) `is`
- (d) `not in`

4. If a, b, and c are integers, which of the following conditions is sufficient to guarantee that the expression `a < c || a < b && !(a == c)` evaluates to true?

- (a) `a < c`
- (b) `a < b`
- (c) `a > b`
- (d) `a == b`

5. Which of the following is the hexadecimal representation of the decimal number `231_{10}`?

- (a) `17_{16}`
- (b) `E4_{16}`
- (c) `E7_{16}`
- (d) `F4_{16}`

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the function `int(x [,base])` is used to convert a string to an integer.
7. True or False: In Python 3, the expression `x >> 1` for `x = 8` evaluates to 3, which is a common misconception about bit shifting.
8. True or False: In Python 3, the expression `['Hi!'] * 4` results in an error.
9. True or False: Python supports automatic garbage collection, which helps manage memory efficiently.
10. True or False: Variable names in Python are not case-sensitive and can be used interchangeably without affecting the code.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the advantages of using binary search over linear search for finding a target value in a sorted list. How does the size of the list impact this choice?
12. Analyze the given code segment that manipulates the variables `r`, `s`, and `t`. Explain the final values of `r` and `s` after execution, and discuss how variable assignment affects output in programming.

End of Paper